

GENERATIVE AI FOR NAMED ENTITY RECOGNITION: PERFORMANCE OF
GPT-4.1-MINI USING LLMNER FOR MIT MOVIE TRIVIA DATA NER

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MSDS 466: Applied Generative AI For Enterprises
June 07, 2026

Abstract

Findings on in-context learning (ICL) performance for named entity recognition (NER) from a 2024 study by Villena et al. are partially replicated, corroborating results on zero-shot, few-shot, and chain-of-thought performance in non-overlapping NER tasks. The MIT movie trivia dataset was used for NER experimentation, and the Python library llmNER, developed by the 2024 study for partial replication, was used. For the selected experiments replicated in this study, the results were consistent: the few-shot, single-turn method performed best on an English dataset. The highest entity-level F1 for a subset of 100 test samples was 68% with few-shot at N=10, and the highest token-level F1 was 85% with few-shot at N=20. However, the scope of this report is far smaller than the scope of the 2024 study. Future work should fully replicate the findings of Villena et al., including experiments with zero-shot, multi-line, and JSON-output schema approaches. ICL for off-the-shelf LLM usage in NER is a promising area of research, and continued exploration of prompting approaches, output schema design, and advanced NER topics is needed.

Introduction

Named entity recognition (NER) is a key natural language processing (NLP) task that involves identifying and extracting labels within text. Recent advances in generative artificial intelligence (AI) technology have opened up new approaches for NER, where off-the-shelf large language models (LLM) can now be employed to extract entities through in-context learning (ICL) (Villena et al. 2024). While success has been documented in using an LLM to extract labels in non-overlapping spans from text with NER, there are many active areas for research to improve performance, including prompt design, in-context learning approaches, output parsing schemas, and more.

This paper iteratively experiments with NER on the MIT movie trivia data, using the llmNER Python library for prompt design, parsing, and model querying. Broadly, this paper seeks to partially replicate the experiments performed by llmNER authors Villena et al. to validate the performance of their library in facilitating zero-shot, few-shot, and multi-turn NER. After exploratory data analysis over the MIT movie trivia data, six experiments are run to test the in-line performance of llmNER and gpt-4.1-mini for NER over twelve non-overlapping entities. The results show that the best entity-level F1 score is 0.668 in the few-shot setting ($N \leq 20$ shots), and the worst is 0.486 in the zero-shot setting. Broadly, these results are consistent with Villena et al.'s findings in their paper, where in-line, single-turn, few-shot approaches on the CoNLL 2003 dataset narrowly outperform other iterations, achieving a $\mu F1$ of 0.735. Further research into the use of libraries like llmNER for non-overlapping generative NER includes experiments across different models, parsing schemas, prompt templates, and datasets. Advanced approaches in NER research may investigate generative AI as a retrieval rather than recognition task for overlapping NER, hierarchical NER, fine-grained NER, and other tasks involving a very large number of entity types (Katz et al. 2023).

Literature Review

NER has historically been performed using rule-based, machine-learning-based, or deep-learning-based approaches (Munnangi 2024). Rule-based NER uses hand-crafted rules to extract entities, but given the manual labor required for annotation, these systems tend to achieve high precision but low recall in experimental settings. Machine learning methods such as conditional random fields (CRFs) and support vector machines (SVMs) aim to mitigate the shortcomings of rule-based methods by learning patterns in text from a training set in a supervised manner. While supervised machine learning methods can achieve high accuracy and

handle ambiguity, they still face shortcomings, including poor out-of-vocabulary generalization and the need for large amounts of annotated data. Deep learning approaches, including Bi-LSTM or Transformer-based models like BERT, deliver even better performance for NER, given their ability to learn complex, nonlinear relationships in the data, though supervised approaches still require large amounts of training data. Pre-trained, encoder-only language models such as BERT have become the standard for NER, but still face computational expense challenges in fine-tuning (Villena et al. 2024).

Off-the-shelf large language models represent a shift in NER, as commercial access to generalized models pre-trained on billions of data points has proven effective, though performance still lags behind state-of-the-art encoder-only models (Liu et al. 2026). One study found that encoder-decoder LLMs such as Mistral and Llama outperform encoder-only models for biomedical NER, but another study found worse performance across benchmark NER datasets (Obeidat and Nahian 2025; Chen et al. 2023). The intention of this paper is to compare varying in-context learning (ICL) methodologies using off-the-shelf LLMs, and it does not seek to compare LLMs to SoTA; however, the author notes LLM limitations in performance, data ethics, and data privacy at this time.

Methods

All experiments are performed in Python 3.9, using a Google Colab Pro notebook environment, using CPU for runtime. Given that the objective of this paper is a partial replication of the experiment done by llmNER authors, some aspects of the 2024 study are kept the same, while others are changed. The codebase and usage of llmNER are carried over from the original experiment to this study. In the interest of time and cost, the number of experiments performed and the OpenAI model selected for experimentation were altered. Only three of the eight

experiments from Villena et al. are directly replicated in this paper: zero-shot + in-line + single-turn, few-shot + in-line + single-turn, and few-shot + in-line + multi-turn. The authors of the 2024 paper did not state the number of shots in their few-shot experiments, so this report presents results for one-shot and few-shot at varying shot counts (N) to compensate. In terms of model selection, only OpenAI GPT-4.1-mini was used in the experiments, whereas the 2024 study used GPT 3.5, Mistral, and Llama. The authors of the 2024 paper experimented on two datasets for English and Spanish NER, but results comparisons are reported only for the English NER dataset.

Six experiments were run to compare in-context learning approaches for off-the-shelf LLM for MIT movie trivia NER. The MIT movie trivia dataset consists of 12 entities: Actor, Plot, Opinion, Award, Year, Genre, Origin, Director, Soundtrack, Relationship, Character_Name, and Quote. The dataset uses Inside-Outside-Beginning (IOB) tagging, where each entity has an associated “B-” and “I-” tag, such as “B-Actor” and “I-Actor,” to delineate entity spans. Each token in the dataset is present as a value in an input array, so sequence tokenization is not required. Tags are identified in a separate array, with tag-to-token mapping determined by array position. A total of 9.4k rows are present in the dataset; however, for the purposes of this experiment, a subset of 100 testing samples, with up to 20 training samples per test sample, is used. Preliminary data analysis is conducted on the data structure to ensure that tag-token pairs are aligned in position and that the entity-type distribution, token count, and token distribution are consistent.

The Python library llmNER is used for prompt development, model querying, and output parsing (Villena et al. 2024). Using llmNER requires initializing a model class, either ZeroShot or FewShot, using an OpenAI model, and initializing the class with a list of possible entities. For

few-shot use, the arguments to the model prediction include an `AnnotatedDocument` object that contains formatted entity spans and ground-truth classifications. All experiments in this paper use in-line parsing of model output rather than JSON formatting, so parsing is performed directly on the model response.

Given the cost and time restraints, the MIT movie trivia dataset was reduced to 100 testing samples. For few-shot runs, it is important that the training samples provided to the LLM include entities in the sequence to be predicted (Liu et al. 2026). For each of the 100 test samples, up to 20 training samples were extracted that contained all entities present in the test sample.

For evaluation, functions that convert `AnnotatedDocument` to spaCy-compatible format were used to run `spacy.scorer` and compute F1, precision, and recall metrics. spaCy scoring is at the entity level, and the entire span identified as an entity in the prediction must match the ground truth span for correctness (iCHAIT 2019). A separate token-level accuracy metric is developed to enable confusion matrix construction. To do this, the spaCy function `offsets_to_bilou_tags` is used to assign BILOU (Beginning, Inside, Last, Unit, Outside) tags to each token in the sequence, and token-level comparison of predictions to ground-truth tags enables clean matrix development.

Experiment 1 uses the `llmNER ZeroShot` model to experiment over zero-shot, in-line, single-turn `gpt-4.1-mini` performance. Experiment 2 repeats experiment 1 design, this time using `llmNER` class `FewShot` for one-shot prompting. Experiment 3 increases the number of shots to 5, experiment 4 increases to 10, and experiment 5 increases to 20. Note that not all testing samples were associated with 20 training samples in the built experimental dataset. In these cases, fewer than N shots were passed to the model, where N is the target number of shots for the experiment.

The author notes that unequal sample counts per test create an imbalance and introduce a new variable to control for; further work should be conducted to better control for this imbalance.

The final experiment adjusted FewShot parameters to enable multi-line prompting. This mode allows for a simple entity to be extracted per turn. In the experiment, the parameter `final_message_with_all_entities=True` is passed so that, in the last turn, the model is prompted to annotate with all possible entities and return them in the same format as single-turn methods.

Results

· Entity-Level Accuracy ---

Experiment #	Experiment Label	# of Samples (Shots)	Precision	Recall	F1 Score
1	Zero-Shot	0	0.585	0.416	0.486
2	One-Shot	1	0.614	0.641	0.627
3	Few-Shot (n=5)	5	0.603	0.638	0.620
4	Few-Shot (n=10)	10	0.665	0.691	0.678
5	Few-Shot (n=20)	20	0.658	0.678	0.668
6	Chain-of-Thought (Multi-Turn)	5	0.663	0.648	0.655

· Token-Level Accuracy (Weighted) ---

Experiment #	Experiment Label	# of Samples (Shots)	Precision	Recall	F1 Score
1	Zero-Shot	0	0.666	0.551	0.468
2	One-Shot	1	0.790	0.736	0.721
3	Few-Shot (n=5)	5	0.826	0.785	0.782
4	Few-Shot (n=10)	10	0.837	0.813	0.813
5	Few-Shot (n=20)	20	0.862	0.843	0.845
6	Chain-of-Thought (Multi-Turn)	5	0.808	0.765	0.756

Figure 1. Entity-level and token-level precision, recall, and F1 across six ICL experiments using llmNER

See Figure 1 above for the results of all six experiments. Token-level metrics generally

outperformed entity-level metrics, suggesting that entity span correctness may be challenging for gpt-4.1-mini with llmNER. At the entity-level, few-shot at N=10 yielded the highest F1, while token-level F2 was highest at N=20. In either case, F1 was higher than in other few-shot experiments by a small margin. F1 for zero-shot performed substantially worse than one-shot or few-shot. Multi-line or chain-of-thought prompting yielded scores comparable to those in few-shot experiments.

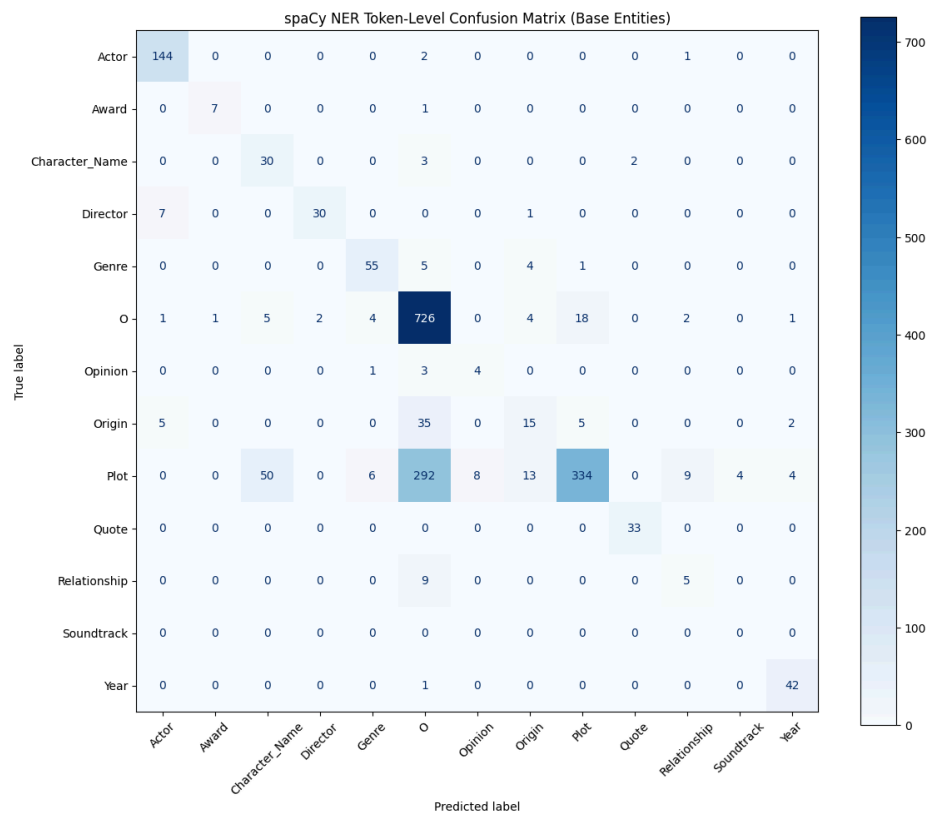


Figure 2: Zero-Shot confusion matrix for token-level NER accuracy

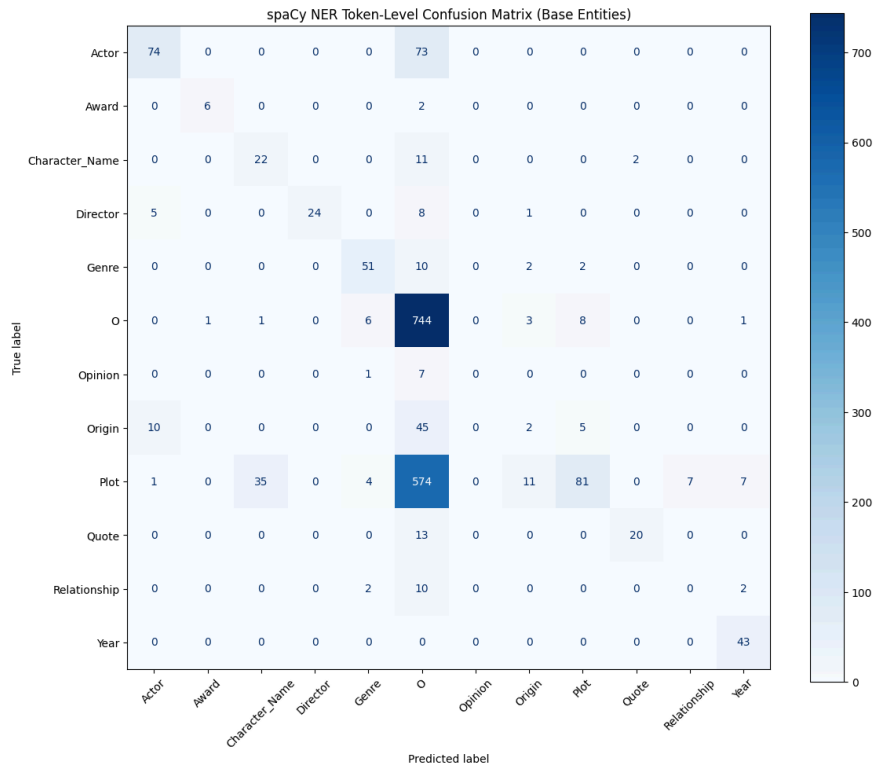


Figure 3: One-Shot confusion matrix for token-level NER accuracy

In reviewing confusion matrices for token-level accuracy, two entities showed near-100% accuracy even in zero-shot settings: “director” and “year” (Figure 2). However, the addition of just a single shot improved accuracy across the board; the “Actor” entity increased in accuracy from near 50% to near 100%, “quote” accuracy increased from 61% to 100%, and “director” accuracy increased from 78% to 97% (Figure 3). Note that the testing set is imbalanced by entity, which may make comparisons of entity-level accuracy less meaningful.

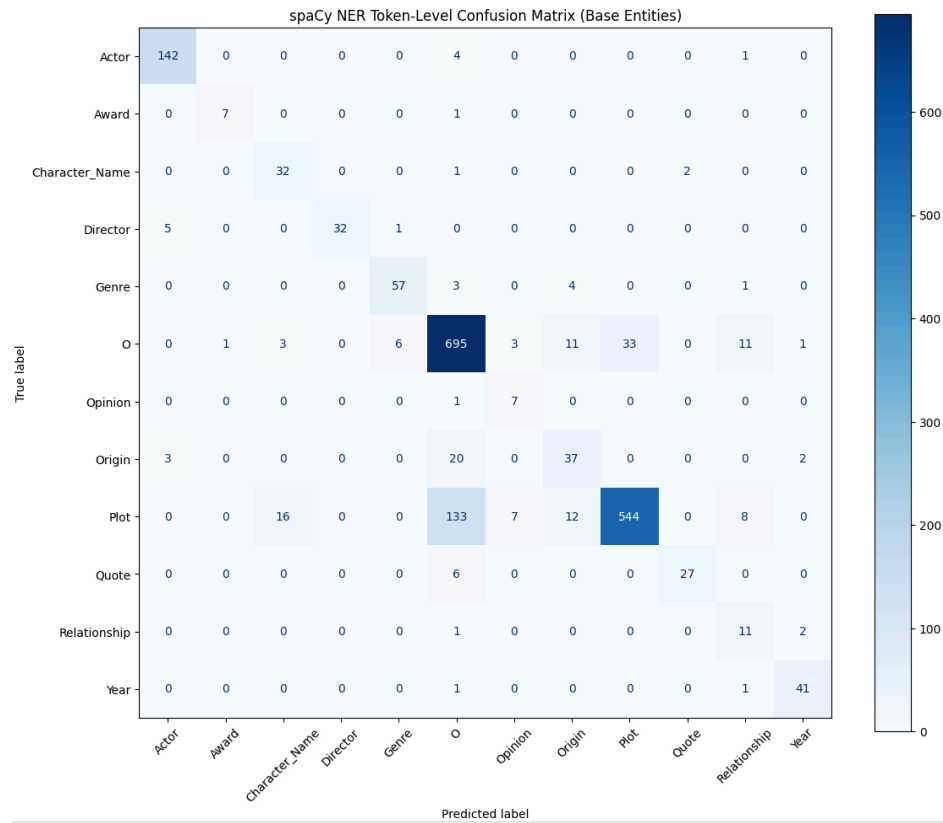


Figure 4: Few-shot, N=20 confusion matrices for token-level NER accuracy

While one-shot classification increased “plot” accuracy to 48%, this entity showed the greatest improvement at few-shot (N=20), achieving around 79% accuracy (Figure 4). Other entities now seeing near 100% accuracy at N=20 include “relationship” and “opinion.” Some entities, however, see a decreased performance at N=20 compared to N=1, including “quote” and “actor”.

Compared with the study for replication, in which llmNER was introduced by Villena et al., some results are consistent, while others require further work to corroborate the findings. This comparison will be limited to Villena et al.’s experiments on the English CoNLL 2003 dataset. For experiments within the scope of this report, Villena et al. found the highest F1 for

few-shot, in-line, single-turn performance over zero-shot single-turn or few-shot multi-line. Their finding is consistent with that of this report. Villena et al. attributed this finding to the inclusion of shots rather than to the prompt template itself. Outside the scope of experiments done in this report, Villena et al. found the best overall performance to be zero-shot, multi-line prompting. Further work is required to fully replicate the experiments conducted in developing Villena et al. and to validate zero-shot, multi-line performance.

Conclusions

Six experiments were run to partially replicate a study by Villena et al. on how in-context learning performs for named entity recognition on MIT movie trivia data, using the llmNER codebase for prompting, model querying, and output parsing. The findings of this report are partially consistent with the study for replication, in that few-shot, single-turn prompting yields the best performance across a selection of other experiments. The highest entity-level F1 for a subset of 100 test samples was 68% with few-shot at N=10, and the highest token-level F1 was 85% with few-shot at N=20. Future work should fully replicate all experiments reported by Villena et al. and include iterations omitted in this paper due to time constraints.

The promise of off-the-shelf LLMs for NER without training data is driving research into ICL approaches and making tooling like llmNER essential to ongoing discovery. To expand upon the work replicated in this report, further work may involve ablation studies to evaluate the impact of prompt templating, model choice, model configuration, and output formatting on performance. Advanced topics in NER, such as hierarchical or fine-grained NER, are also worth exploring in generative AI settings, both in experiments with ICL methods and in comparisons with SOTA methods (Katz et al. 2023).

References

- Chen, Xuanting, Junjie Ye, Can Zu, Nuo Xu, Rui Zheng, Minlong Peng, Jie Zhou, Tao Gui, Qi Zhang, and Xuanjing Huang. 2023. "How Robust is GPT-3.5 to Predecessors? A Comprehensive Study on Language Understanding Tasks." *ArXiv* abs/2303.00293.
- iCHAIT. 2019. "How Is the Overall F Score of an NER Model Calculated?" *GitHub Issue #4094*, *spaCy Repository*, August 7, 2019.
<https://github.com/explosion/spaCy/issues/4094>
- Katz, Uri, Matan Vetzler, Amir Cohen, and Yoav Goldberg. 2023. "NERRetrieve: Dataset for Next Generation Named Entity Recognition and Retrieval." In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 3340–3354. Singapore: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.findings-emnlp.218>
- Liu, Zhaoli, Tao Qin, Bohao Liu, Qindong Sun, and Shancang Li. 2026. "Knowledge Enhanced Prompting for Few-shot Named Entity Recognition". *International Journal of Machine Learning and Cybernetics* 17 (2). <https://doi.org/10.1007/s13042-025-02902-8>.
- Munnangi, Monica. 2024. "A Brief History of Named Entity Recognition." *arXiv* preprint arXiv:2411.05057. <https://arxiv.org/abs/2411.05057>
- Obeidat, M. S., M. S. Al Nahian, and R. Kavuluru. 2025. "Do LLMs Surpass Encoders for Biomedical NER?" *Proc (IEEE Int Conf Healthc Inform)* 2025: 352-358.
<https://doi.org/10.1109/ichi64645.2025.00048>.
- T-NER. n.d. *MIT Movie Trivia*. Hugging Face Datasets.
https://huggingface.co/datasets/tner/mit_movie_trivia
- Villena, Fabián, Luis Miranda, and Claudio Aracena. 2024. "llmNER: (Zero|Few)-Shot Named Entity Recognition, Exploiting the Power of Large Language Models." *arXiv* preprint arXiv:2406.04528. <https://arxiv.org/abs/2406.04528>